DESCRIPTION OF COURSEWORK

Course Code	EEE407
Course Name	Electrical Energy Utilization
Lecturer	Dr. Ng Zi Neng
Academic Session	2024/09
Assessment Title	Assignment (group)

A. Introduction/ Situation/ Background Information

Energy is a critical driver of a country's development and growth. In Malaysia, the government has implemented comprehensive national energy policies and regulations to promote sustainability and foster growth within the energy sector. Numerous initiatives have been introduced to encourage the adoption of advanced technologies by both companies and consumers, enabling more efficient electricity use. Given the sector's central role in socioeconomic progress, it is imperative that Malaysia's energy infrastructure remains resilient and adaptable to both domestic and global trends. Ensuring the sector's sustainability and competitiveness is key to maintaining long-term growth.

For energy consumers, efficient use of electrical energy involves minimizing waste while maintaining the same level of comfort or productivity with reduced energy input. The first step in achieving this is conducting an energy audit in accordance with established standards, which quantifies a building's electrical energy usage. This process includes analyzing consumption by specific areas or equipment to identify sources of inefficiency. Additionally, the audit evaluates the energy performance of individual systems or equipment using precise measurement tools to ensure that all components are operating optimally.

A Registered Electrical Energy Manager (REEM) is qualified to carry out energy audits and recommend Energy Conservation Measures (ECMs) in accordance with industry standards and best practices. The primary objective of an energy audit is to provide actionable insights and strategies to help clients reduce energy consumption and associated costs. ECM recommendations should span a range of investment options, allowing clients to make informed decisions based on

a thorough analysis of return on investment (ROI) and payback periods, as outlined in the audit report.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Explain the important aspects of efficient electrical energy utilization.

CLO 3 Evaluate energy saving solution based on given energy data.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

A group of 4 – 5 members is set to perform an energy audit on a low/medium voltage commercial building (Tariff B and Tariff C) or a low voltage industrial building (Tariff D). The goal is:

a. To investigate and analyze the electrical energy consumption of the premise.
b. To recommend ECMs in all ranges of costs: zero to high-cost measures.
c. Collaborate effectively as a team and apply professional engineering practices when conducting energy audits.

CLO 1 is mapped to PLO10, Communication. Therefore, it is important to apply the following methodologies:

a. Communicate with the owner of the business and arrange an on-site visit to the business premise.
b. Basic field measurement with available/inexpensive tools, i.e., lux meter app on smartphone, thermometer, digital clamp meter/multimeter (if available), etc.
c. Data collection through readings of sub-meters, paper energy bills, technical documents, and interview with staffs.
d. Analyze data obtained and present findings through report and group presentation.

To complete this assignment, students need to communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. The general guideline for the assignment report is shown in Part F.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Energy audit report	20
2.	Group presentation	10
	TOTAL	**30**

F. Task(s)

Report

Write a report of an energy audit based on the selected business premise. The report should include but not limited to the following sections:

No	Section	Contents
1	Introduction	• Highlight on current energy scenario and the needs for energy saving. • Government initiatives towards energy efficiency. • Objectives, scope, and type of audit. • Key systems and equipment audited.
2	Background of the Company	Background of the company/ organization/ building • Information of the company, when it is established, function of the company, the owner, etc. • Describe the building, include information that are crucial to energy consumption i.e., location and weather (name of city or town), types of property (landed or high rise), number of floors/rooms, floor area, number of doors/windows, number of employees/occupants, age of building, hours of operation etc. • Brief description of services in the buildings. • Process flow diagram and major components of operations. Management of Energy (if available) • Policy and targets. • Energy data, documentation and monitoring. • Compliance with towards the regulations. • Internal energy management team. • Internal energy audit team.
3	Electrical Energy Audit	• Plans/timeline of the audit activity, date of audit, name of auditees and their duties. • Desktop data collection such as facility drawings, technical specification sheet of equipment, bills, etc. • Total electricity consumption trends. Show bills with monthly energy consumption, tariff, and cost. • Field data collection via meter readings and simple measurement tools. Use an audit checklist such as Lighting data form, you may come up with your own form or use an existing template available from the internet. • Analysis of energy consumption by equipment/system. • Analysis of energy consumption by areas/levels.

		• Remarks and conclusions from consumption profiles and trends from energy bills and measurements. Suitability of energy supply types and tariff/prices. • Current energy performance status (energy intensity) for building, processes and major energy using equipment.
4	ECM recommendations and potential savings	• Recommendation of 3 zero cost and 3 low cost: housekeeping, minor repairs. • Recommendation of 3 medium cost and 3 high cost/major investment required: major repair/replacement/introduction of new technologies. • For all ECMs, estimate potential of energy savings in energy units and currency and show calculation methods. • For medium and high cost ECMs, determine the potential returns from the costs to implement energy saving measures (Simple Payback Period/Investment Rate Return). • Guides on how to implement proposed energy saving measures. Proposed action plan and estimated time required to implement each measure. • Financing options/Government Incentives available. Describe how the company could finance the implementation of the recommended ESMs (internal or external funds). • Proposed types of relevant non-dependent variables and their correlation to energy consumption to determine the baseline energy. • Develop the linear model using regression method.
5	Summary	• Summary of recommendations, estimated annual kWh and estimated cost savings, estimated investment cost and payback (in table form).
6	Reference	• List down the references used.
7	Appendices	• Electricity bills. • Measured energy data. • Photos from site assessment. • References such as Standards documents and best practices.
8	Report Structure	• Well-presented report with consistent formatting.

The format of the report has to be consistent:

- Use the Assignment cover page available in Moodle.
- Use Times New Roman font with size 12 and 1.5 line spacing.

- Number all main topic (**bold**) and sub-topic (***bold and Italic***).
- Figures and tables have to be captioned and placed according to the standard format, i.e., Figure 1.1 “captions” is placed after the figure and Table 1.1 “captions” is placed before the table.
- Cite references with APA citation format.
- A minimum word count of 6000 words from Introduction to Conclusion.
- Tabulate and organize all data points and variables.
- Plot all relevant graphs/charts using proper software i.e., MS Excel, MATLAB, and other mathematical software.
- You are also encouraged to use any open-source 2D or 3D design software to draw specific room layout and placement of appliances i.e., RoomSketcher | Create Floor Plans and Home Designs Online
- Keep technical descriptions comprehensive and concise. Consider that in most cases decision makers are not necessarily engineers.

Presentation

For group presentation, all members of the group are required to present their work with interesting slides, strongly recommended to show pictures and videos during the audit activity. Presentation should not be longer than 15 minutes.

Submission

Each group will need to submit a report in pdf, upload it to Moodle Assignment submission tab. The report submission is due on Week 10, followed by the group presentation on Week 11. Exact date will be announced later.

APPENDIX 1

MARKING RUBRICS

Component Title	**Energy audit report**				**Percentage (%)**	**20**	
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Introduction [CLO1-PLO10:A3]	The introduction clearly identifies the background of current issues and relates to the government's initiatives with strong justification to the importance of current engineering problem. The objectives are clear.	The introduction clearly describes the background of current issues with the government's initiatives, but lack the justification to the importance of current engineering problem. The objectives are clear.	The introduction adequately describes the background of current issues but some government's initiatives are not relevant. The objectives could be clearer.	Some current issues are not clearly described, the government's initiatives are not relevant, the objectives are not clear.	Most topics are not relevant. Objectives are not clear.	**10**	
Background of company and energy audit – Data collection [CLO1-PLO10:A3]	Procedures of data collection are clearly proposed. Details and layout of the building are clearly explained. Information of all appliances/equipment are reported and tabulated, supported by picture proofs of current appliances/equipment placement/manuals. Daily/ Monthly energy profile and monthly energy cost are presented using appropriate charts/plots, picture proofs are provided. Field and desktop data are provided.	Procedures of data collection are clearly proposed. Details and layout of building are clearly explained. Information of most appliances/equipment are reported and tabulated, supported by picture proofs of current appliances/ equipment placement/ manuals. Daily/ Monthly energy profile and monthly energy cost are presented using appropriate charts/plots, picture proofs are provided. Only desktop data are provided.	Procedures of data collection are proposed. Details of building are briefly explained. Information of most appliances/ equipment are reported and tabulated, but some are not supported by picture proofs/ manuals. Daily/ Monthly energy profile and monthly energy cost are presented using appropriate charts/ plots, picture proofs are provided. Only desktop data are provided.	Details of building and data collection procedure are vaguely explained. Information of most appliances/ equipment are not presented clearly and most are not supported by picture proofs/manuals. Incomplete monthly/ daily energy data collection.	No/Unclear details of building and data collection procedure. Information of appliances/ equipment are not well reported and not supported by any picture proofs/manual. Incomplete monthly/daily energy data collection.	**30**	
Energy audit – Analysis of energy consumption [CLO3-PLO2:C5]	Energy usage is apportioned according to area and types of appliances, well reported with appropriate charts and plots. Estimation of energy usage by each area/appliance is clearly demonstrated and agrees well with field data and daily/monthly usage by utility provider. Improvement on estimation is suggested. Details of each chart are clearly analyzed and discussed. Excellent explanations on the relationship between energy usage and area/appliance with relevant theories provided.	Energy usage is apportioned according to area and types of appliances, well reported with appropriate charts and plots. Estimation of energy usage by each area/appliance is clearly demonstrated and agrees well with daily/monthly usage by utility provider. Improvement on estimation is suggested. Details of each chart are clearly analyzed and discussed with convincing explanations on the relationship between energy usage and area/appliance.	Energy usage is apportioned according to area and types of appliances, well reported with appropriate charts and plots. Estimation of energy usage by each area/appliance is briefly demonstrated. Charts are analyzed and discussed with sufficient explanations on the relationship between energy usage and area/appliance.	Energy usage is apportioned according to either area or types of appliances only. No demonstration on energy usage estimation. Some data are not presented using charts and plots. Details of each result could be clearer. Poorly written/ incomplete explanations of the relationship involved. Discussion shows little understanding of the results.	No energy usage apportioning and estimation of energy usage. Unclear details on results, poorly discussed with no appropriate charts and plots.	**20**	

ECM recommendations and potential savings [CLO3-PLO2:C5]	All 3 ECMs from each category provided are suitable and clearly justified. All calculations on potential savings and payback periods are clearly demonstrated with proper assumption and price of new appliances/ equipment quoted from reliable source. Action plan and timeline are proposed. Financing options/ Government Incentives are provided. Linear model using regression method is developed.	All 3 ECMs from each category provided are suitable and clearly justified. Most calculations on potential savings and payback periods are demonstrated but some lack verified pricing. Action plan and timeline are proposed. Financing options/ Government Incentives are provided.	Some ECMs provided are not suitable/ need clearer justification. Most calculations on potential savings and payback periods are demonstrated but some but some lack verified pricing. Action plan and timeline are proposed.	Some ECMs provided are not suitable/ not well justified/ not enough as per report guideline. Some calculations on potential savings and payback period are unclear/ incorrect. No verified pricing.	Some ECMs provided are not suitable/ not well justified/ not enough as per report guideline. Calculation of potential savings are not done.	**20**	
Summary and references [CLO1-PLO10:A3]	The objectives of the report have been achieved, all key results, analysis, recommended ECMs, and potential savings are summarized in a concise manner. Complete references to any book, articles and websites indicating in-text citations report with correct referencing format are in place.	The objectives of the report have been achieved. Most key results, analysis, recommended ECMs, and potential savings are summarized. Proper references are used with correct referencing format are in place.	Most objectives of the report have been achieved, only some key points are summarized. Proper references are used with correct referencing format are in place.	Some objectives are not achieved, key points are not well summarized. More references needed. Correct referencing format are in place.	Objectives are not achieved. More references needed. Incorrect referencing format.	**5**	
Appendices and figures [CLO1-PLO10:A3]	Many relevant on-site pictures. Clear figures, captioned, titled, and labelled accordingly and all are mentioned in the main content.	Many relevant on-site pictures. Clear figures, captioned, titled, and labelled accordingly and most are mentioned in the main content.	Sufficient relevant on-site pictures. Most figures are clear, some are not captioned, titled, and labelled accordingly and some are not mentioned in the main content.	Sufficient relevant on-site pictures. Poor quality figures, most are not captioned, titled, and labelled accordingly and not mentioned in the main content.	Little or no relevant on-site pictures. No caption and not mentioned in the main content.	**10**	
Format [CLO1-PLO10:A3]	All the sections are presented in a neat and tidy manner with consistent format and style as per report guidelines. All figures and tables are clear and properly labelled and captioned.	Most sections are presented in a neat and tidy manner with consistent format and style as per report guidelines. Most figures and tables are clear and all are properly labelled and captioned.	Some sections are inconsistent in the format and style and did not follow some of the report guideline. Most figures and tables are clear and some are not properly labelled and captioned.	Many sections are inconsistent in the format and style and did not follow the some of the report guideline. Some figures and tables are blurry and some are not properly labelled and captioned.	Messy report and difficult to read with inconsistent format and style. Most figures and tables are blurry and not properly labelled and captioned.	**5**	
					TOTAL	**100**	

MARKING RUBRICS

Component Title	Presentation [CLO1-PLO10:A3]				Percentage (%)	10	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Content of slides • Introduction and background	Excellent introduction and objective of this audit, and descriptions of the business premise are presented. Selection of business premise to perform energy audit is well justified.	Proper introduction and objective of this audit, and descriptions of the business premise are presented. Good selection of business premise to perform energy audit.	Introduction and objective of this audit, and descriptions of the business premise are well-presented.	Introduction and objective of this audit are presented. The descriptions of the business premise are not clear.	Lacks proper introduction of this audit and descriptions of the business premise.	**10**	
Content of slides • Analysis of energy consumption • ECM recommendation and justification • Additional materials: pictures of audit process, field measurement results	Covers all key findings from the audit. Explain the contents in a comprehensive manner. Well-thought ideas were presented to the audience with effective and persuasive ECM recommendations , return on investment and simple payback period. Audit process well recorded with pictures and field measurements.	Covers most key findings from the audit. Contents are well-explained. Able to convey ideas to the audience with effective and persuasive ECM recommendations , return on investment and simple payback period. Audit process well recorded with pictures and field measurements.	Covers some key findings from the audit, more important information could be presented to instill confidence of the audience. Some ECM recommendations , return on investment and simple payback period are presented.	Covers few findings from the audit, lack some information that may otherwise provide a clearer picture to the viewer. More important information could be presented to instill confidence of the audience.	Lacks many key information from the audit. Overall, not persuasive in conveying ideas to the audience.	**30**	
Creativity of slides	Well-designed with good use of figures and videos, presented key data and results with interesting charts and figures. Key points are used instead of just text.	Well-thought design, presented key data and results with interesting charts and figures. Some slides are quite packed with texts or irrelevant images.	Average design, presented key data and results with charts and figures. Quite packed with texts or irrelevant images.	Quite packed with texts or irrelevant images, most data are presented via text. Summary of key data could be done better.	Mostly texts and minimum effort is put into designing the slides.	**20**	
Delivery	Smooth delivery that captures audience attention. Duration of presentation is good and not exceeding the 15 min mark.	Good delivery that holds audience attention most of the time. Time management is good.	Delivery of information that was able to hold audience attention most of the time. Time management can be improved.	Able to capture audience's attention once awhile. Presentation is either too lengthy or too short.	Delivery not smooth, and audience attention is often lost. Presentation is either too lengthy or too short.	**30**	
Q&A	Clearly, accurately and professionally answers and defends audit findings based on	Accurately answers and defends audit findings based on theoretical concepts.	Able to answer and defend the audit findings but sometimes not able to clearly justify the	Able to partially answer and defend the audit findings but have difficulty to justify them with	Have difficulty understanding the question hence giving accurate answers.	**10**	

	theoretical concepts.		selection of analysis method.	theoretical concepts.			
					Total	**100**	

Note to students: Please print out and attach this appendix together with the submission of coursework.